

Om Shrivastava

Junior Undergraduate

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021-2025	B.Tech-Electrical Engineering	Indian Institute of Technology Kanpur	8.8/10
2021	CBSE(XII)	Shiv Jyoti International School,Kota	89.2%
2019	CBSE(X)	K.John Public School, Nagpur	94.6%

Scholastic Achievements

- Secured **Gold medal** in **Inter-IIT Tech Meet 11.0** for the problem statement by Jaguar and Land Rover(JLR). 2023
- Secured **All India Rank 1282** in JEE Advanced & **All India Rank 2758** in JEE Mains among 1 Million candidates.2021
- Selected among **top 300 participants** for Quant Trading Winter School, organized by **Optiver & IIT Delhi**. 2023
- Attained **Gold Level** in World Quant IQC and was offered the **Brain quantitative research consultant** position. 2023

Work Experience

Industry 4.0 | Prof. Nalinaksh S. Vyas | Summer Intern | IIT Kanpur (May'23 - Present)

Objective	• Simulate, implement, and drive the adoption of Industry 4.0 principles and technologies in a factory setting.
Approach	• Implemented Multilabel classification using transfer learning for defect detection and achieved 100% accuracy . • Devised scheduling algorithm on Python , optimizing factory production with inventory data & client needs. • Executed Real-Time Machine Monitoring using data from NI-OPC servers on LabView for 15+ machines .
Impact	• Enhanced efficiency by 17% through optimized machine scheduling and monitoring, driving productivity.

Key Projects

MUS Rover: Self learning Musical compositional Rules | Prof. Vipul Arora | IIT Kanpur (May'23 - Present)

Objective	• To research the possible way to build a model to extract the best compositional rule from the input audio.
Approach	• Conducted extensive literature review to find the possibilities to extract the best features from input audio. • Utilized Probability theory, Information theory, and Machine learning to Solve optimization problems. • Implemented signal processing code to extract the pitch from a dataset of classical music for creating test data
Results	• The model produced the best feature distribution and the probability of all the features it extracted.

Jaguar Land Rover's Robotic Charging Challenge | Inter IIT Tech Meet 11.0 | 🏆 (Jan'23-Feb'23)

Objective	• Develop a robotic arm for automatically plugging the socket into the charging port of a JLR Electric Vehicle .
Approach	• Trained Yolov5 model for detection of a charging port in obstructed views with the help of a custom dataset. • Used Transfer Learning based model for pose estimation and trained YOLOv5 model to detect CCS2 port.
Result	• Won Gold medal & achieved best arm design & simulation among 19 IITs within budget worth INR 80,000 .

Computer Vision & Sensors of Automi | Mentor:Prof. Venkatesh | Team Humanoid | 🏆 (Dec'22)

- Implemented **Hough transformation** to detect the center of a revolving archery board & calculate it's angular velocity
- Tested and used **FSR-402** and **MPU-6050** with Arduino, which could be used for terrain detection while walking

Grasping | Mentor:Prof. Venkatesh | Team Humanoid (Jan'23-Mar'23)

- Performed **Torque analysis** for **estimation of load on the actuator** and calculated the parameter to get required torque
- Applied **Stereo-Block Matching algorithm** and **YOLO-V5** for detection of objects, their depth, and dimensions

Desktopography | Project Mentor | Electronics Club, IIT Kanpur | 🏆 (May'23-Jul'23)

- Led a team of **9 members** to implement a **touch feature** on projection using a depth map generated from Intel RealSense
- Generated a **mapping** between projection and computer screen based on detected projection size and computer resolution
- Guided computer to click at mapped position using **PyAutoGUI library** based on a finger lying below a threshold distance.

Technical Skills

Programming Languages	Libraries	Software
C,C++, La T _E X,python	Open CV,Numpy,Pandas,Keras,Matplotlib	Arduino,Edge Impulse

Relevant Courses

(*:Ongoing)

Fundamentals of Computing	Introduction to Machine Learning*	Introduction to Electronics
Signal systems & network	Linear Algebra	Data structure & Algorithms*

Positions of Responsibility

Coordinator, Electronics Club (Apr'23-Present)

- Leading a **two-tier** team structure comprising of **25 secretaries** to foster culture in the domain of electronics.
- **Mentored 3 projects** Based on electronics, BCI, and Computer vision in the summers comprising **35+ mentees**.
- Managed the budget Worth **INR 1.65 Lakh** which was presented to the council and approved by the Senate.

Technical Team Member, Team Humanoid (Sep'22-Present)

- Implemented **computer vision** and **Image processing** algorithms to complete tasks based on Stereovision for the bot.
- Worked with a team **comprising 20 members** of various domains to redesign and optimize bot electrical architecture.
- Finished project on Grasping, Computer vision and currently preparing for the **IROS competition** organized by **IEEE**.